Conjecture 00000000684: a classical circle has an infinite finitely generated group

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1 The standard virtual group and the counterexample

We use the standard Kauffman group $G(D)$ of a virtual diagram, defined by its Wirtinger presentation. The source names a virtual diagram group without further specifying a variant. This standard model is defined in Kauffman, *Virtual Knot Theory*, Europ. J. Combinatorics 20 (1999), Section 4, printed pp.671–672: generators correspond to arcs between classical undercrossings, and relations to classical crossings. Virtual crossings contribute no new generators or relations. At printed p.663 the author explicitly uses "trivial fundamental group" to mean the infinite cyclic group $\mathbf{Z}$.

Take the actual round circle

$$S^1 = \{z \in \mathbf{C} : |z| = 1\}, \qquad K : S^1 \longrightarrow \mathbf{C} \times \mathbf{R} = \mathbf{R}^3, \quad K(z) = (z, 0).$$

Its planar projection is $z \mapsto z$. It is an actual embedding, with no self-crossings and no virtual crossings. Thus it gives a classical diagram of the ordinary round unknot. Its standard Wirtinger group has one arc generator and no crossing relations, so is isomorphic to $\mathbf{Z}$. In particular it is finitely generated and infinite, although this knot is classical. This contradicts the forward implication of the proposed equivalence. An additional Gauss-diagram criterion cannot repair that false necessary implication.

2 Actual crossings and arcs, rather than an assigned presentation

The formal source uses Mathlib's actual complex unit circle. It first considers a crossing-free diagram given by a continuous injective map $f : S^1 \to \mathbf{C}$. The set of crossing preimages is defined from the actual map:

$$U_f = \{x \in S^1 : \exists y \neq x,\ f(y) = f(x)\}.$$

Injectivity proves $U_f = \varnothing$. Every possible undercrossing preimage subset is therefore empty as well. This avoids treating overcrossing preimages as undercrossings in a general diagram; the two sets are both empty in this subclass.

Arcs are the genuine connected components of $S^1 \setminus U_f$. An explicit homeomorphism identifies this actual complement with S^1, and the connectedness of the circle proves that its component type has exactly one element. The formal generator type is this component type, not a separately assigned one-element label.

The classical-crossing index type is also defined from actual pairs of distinct circle points with equal planar image. It is proved empty. A crossing-relator assignment has this type as its domain, and its range is consequently empty. Lean proves this for *every* relation assignment on the actual empty crossing type. Thus no choice of signs, underpassing branch or usual Wirtinger convention can add a crossing relation in this example. There is no need to encode the crossing formula for nonempty diagrams to settle this crossing-free case.

The diagram group is the actual `PresentedGroup` on those arc generators and the crossing-indexed relator set. Lean proves that its quotient map from the free group is injective as well as surjective, yielding the genuine isomorphisms

$$G(D) \simeq F(\text{one actual arc}) \simeq (\mathbf{Z}, +).$$

Mathlib's actual group finite-generation property and infinitude are then proved for this presented group.

3 Classical realization and the exact scope

For the round diagram, the projection is the actual subtype inclusion $S^1 \hookrightarrow \mathbf{C}$. The formal source also constructs the actual spatial embedding $z \mapsto (f(z), 0)$ and checks that its projection is f and its height is zero. For $f(z) = z$ it additionally records the unit-norm equation, binding the construction to the ordinary round circle.

The predicate `ClassicalRealization` records this sufficient classical realization for the crossing-free subclass. It is not presented as a classification or definition of arbitrary virtual knots. In particular, it does not replace nonclassicality for arbitrary diagrams by failure to embed in a plane. The standard round diagram already has no virtual crossings, so its classical status requires no equivalence or unknot-recognition algorithm.

The final formal statements refute both the restricted equivalence and, separately, its necessary implication: an infinite finitely generated diagram group would prohibit such an explicit classical realization. The actual round diagram satisfies the group properties and has that realization, so this implication fails.

We use Kauffman's named standard Wirtinger model. The Lean source does not reprove invariance under every extended Reidemeister move, the identification with the fundamental group of a complement, or a theorem about every alternative extended virtual-group definition. None of these additional results is used to assign a group to an unrelated presentation: the arc and crossing types are derived from the actual circle projection before forming the group.

4 Verification

Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`

are pinned. The package includes a new root build, transitive axiom audits and full official `leanchecker --fresh Main` replay. No `sorry`, `native_decide` or extra axiom is used. OpenAI Codex assisted; a separate supervising Codex agent reviewed source, report and PDF. This is AI review, not human certification. Organizer acceptance is pending.